7 The Doppler effect

7.1 Introduction

You may have experienced this: A car travelling in the opposite direction on the highway passes by you, and you notice that the pitch of the sound of the car's engine drops appreciably as it passes.

In fact, if you were able to compare the pitch you hear with the pitch that the driver of the passing car hears, you would realize that your perceived pitch is higher while the other car is approaching you, and lower while the other car is receding away from you.

All of this means that the observed pitch, or the frequency of a sound, depends on how the emitter and the observer are moving relative to each other. This is known as the **Doppler effect**, after Christian Doppler (1803–1853), who first studied the effect, including setting up an experiment where two groups of trumpeters with perfect pitch, one at a railway station, and the other in a passing railway car, played the same note at the same time, whereupon a significant frequency difference between the two groups was heard. Doppler predicted that the same effect would apply to "ether waves", and astronomers would be able to apply it to measure velocities of stars, a remarkable prediction which we now know to be true. One of the most spectacular applications of the Doppler effect is to measure the velocities of distant galaxies relative to our own, which allows us to measure the rate of expansion of the universe.

In this chapter, we will derive an expression for the change in frequency in a sound signal, when the source and the observer are moving relative to each other.

7.2 Deriving the Doppler formula

Imagine the following scenario: A source of sound S and an observer O are both moving along the x axis towards each other. S is emitting sound of frequency f and moving in the $+x$ direction, while O is moving in the $-x$ direction. We will follow two successive wave fronts: Wave front number 1 is emitted at $t = 0$, and number 2 is emitted at

$t = T$, the period of the wave according to S. We will calculate at what times t_1 and t_2 they reach O. The difference $t_2 - t_1 = T'$ is the period of the sound according to O. What we want to do is find the relation between T and T'.

Let's write expressions for the position as a function of time for each of the four characters in the story. First, the source of sound:

$$x_S(t) = x_{S,i} + v_S t \tag{7.1}$$

where $x_{S,i}$ is the position of S at $t = 0$ and $+v_S$ is the velocity of S along the x axis.

Next, we do the same for the observer. O's velocity must be negative as O and S are moving towards each other, so we will write it as $-v_O$:

$$x_O(t) = x_{O,i} - v_O t \tag{7.2}$$

Wave front #1 is emitted at $t = 0$, when $x_S = x_{S,i}$, and it propagates with the speed of sound v, so its position as a function of time is

$$x_1(t) = x_{S,i} + vt \tag{7.3}$$

Wave front #2 is emitted at $t = T$, when $x_S = x_{S,i} + v_S t$, so its position as a function of time is

$$x_2 = x_S(T) + v(t - T) = x_{S,i} - (v - v_S)T + vt \tag{7.4}$$

At $t = t_1$, wave front #1 reaches the observer, so $x_1(t_1) = x_O(t_1)$:

$$x_{S,i}t_1 + vt_1 = x_{O,i} - v_O t_1 \tag{7.5}$$

which we can write as

$$x_{S,i} - x_{O,i} + (v + v_O)t_1 = 0 \tag{7.6}$$

Similarly, at $t = t_2$ wave front #2 reaches the observer, so $x_2(t_2) = x_O(t_2)$, and therefore

$$x_{S,i} - (v - v_S)T + vt_2 = x_{O,i} - v_O t_2 \tag{7.7}$$

which we can write as

$$x_{S,i} - x_{O,i} + (v + v_O)t_2 = (v - v_S)T \tag{7.8}$$

Now if we subtract (7.6) from (7.8), and set $t_2 - t_1 = T'$, we find

$$(v + v_O)T' = (v - v_S)T \tag{7.9}$$

or, in terms of the frequencies $f = 1/T$ and $f' = 1/T'$,

$$f' = f\frac{v + v_O}{v - v_S} \tag{7.10}$$

Equation (7.10) applies to a particular scenario, namely when the source and the observer are both moving towards each other. To generalize the formula, we have to consider the alternatives. By retracing the steps, we would find that

- if O is moving *away* from S, the sign in the numerator should be a minus; and
- if S is moving *away* from O, the sign in the denominator should be a plus.

So the general Doppler formula is:

$$f' = f\frac{v \pm v_O}{v \mp v_S} \tag{7.11}$$

where the top sign is chosen in the numerator if O is moving towards S, and the top sign is chosen in the denominator if S is moving towards O.

Another way of remembering the signs is that the perceived frequency increases in if either O or S is moving towards the other.

Example 7.1: *A bat pursuing a stationary insect flies in a straight line towards the insect with speed v_{bat}. If the bat emits sound of frequency f, what is the frequency f' heard by the bug? If the bug reflects that frequency back towards the bat, what is the frequency f'' detected by the bat? Derive a formula to calculate the velocity of the bat from the value of $(f'' - f)/f$ (call this x).*

The frequency heard by the insect is

$$f' = f\frac{v}{v - v_{bat}}$$

since the bug (O) is stationary, and the bat (S) is moving towards the bug.

The frequency detected by the bat is

$$f'' = f'\frac{v + v_{bat}}{v}$$

since now the bat (O) is moving towards the bug (S), and the bug is stationary.

Altogether,

$$f'' = f\frac{v + v_{bat}}{v - v_{bat}}$$

From this, after a little algebra we get

$$v_{bat} = v\frac{x}{2 + x}$$

7.3 *The effect of wind*

What if the air itself is moving (relative to the ground)? In that case, we may reason as follows: In still air, the wind carries the waves to the listener with speed v. If there is a wind of speed w *towards* the listener, then the speed with which the waves travel towards the listener is $u = v + w$. But if the wind blows *away* from the listener, then the speed with which the waves travel towards the listener is $u = v - w$.

Either way, we have to substitute u for v in the Doppler formula, which thus becomes

$$f' = f\frac{u \pm v_O}{u \mp v_S} \tag{7.12}$$